Hongru Yan

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EDUCATION

Tsinghua University Beijing, China

Bachelor in Mathematics and Physics+Measurement Control Technology and Instruments 2021.09-2025.07

- **GPA**:3.95 (Rank 2, with 8 courses A+ and 20 courses A)
- **Major Courses**: Programming Fundamentals (A 99), Fundamentals of Physics (1) (A+ 100), Mathematical Physics Equations (A+ 100), Data Structures and Algorithms (A+ 100), Design and Practice of Circuit System (A+ 100), Probability Theory (A 98), Modern Operation System(A 98), Introduction to Deep Learning(A+ 100)
- **Awards**:
  National Scholarship (top scholarship of undergraduates in China, 0.2%) Ministry of Education
  Academic Excellence Scholarship Tsinghua University

ACADEMIC RESEARCH

Microsoft Research Asia Beijing, China

Research Intern 2025.04-Present

- Working on Interactive World Models on Real World Experiments with Physical Rules.

Research Projects at UCSD

Gaussian Swaying: A Surface-Oriented Framework for Wind-Driven Dynamics with 3D Gaussians

Instructed by Prof. Zhuowen Tu, Department of Cognitive Science at University of California, San Diego 2024.09-2024.11

- Explored advanced Material Point Method(MPM) for more efficient and realistic simulation
- Propose Gaussian Swaying, a novel framework for physics-based wind simulation with 3D Gaussians, leveraging its potential for real-time rendering.
- Propose Surface-oriented Material Point Method (S-MPM) that treats Gaussians as surface representations, which efficiently bridges dynamic simulation and rendering.
- Creating compelling wind-drvien dynamic simulations with shading effects in real-time.
- Complete the project and submit the paper to CVPR as the first author, including coming up with the idea, conducting the experiments, writing the paper and drawing the figures.

Gaussian-Det: Learning Closed-surface Gaussians for 3D Object Detection

Instructed by Prof. Yueqi Duan, Department of Electronic Engineering at Tsinghua University 2024.01-2024.05

- Utilized Gaussian splatting as scene representation for object detection with high accuracy and efficiency in an elegant way
- Proposed the use of closure as a major indicator of detection, balancing simplicity and correctness
- Proposed a framework named Closure Inference Module(CIM) to infer closure to find geometry clues in 3D object detection
- Completed the overall structure, wrote code, produced datasets, and accomplished the project paper.

ShadowNeRF: Learning Neural Radiance Field with Sight Degradation and Recovery

Instructed by Prof. Yueqi Duan, Department of Electronic Engineering at Tsinghua University 2023.09-2023.11

- Proposed a method for sight degradation and recovery in neural rendering through pre-training and fine-tuning.
- Contributed to the idea, illustrations, code development, and experiment validation.
- Experimentally validated the effectiveness of the proposed method in downstream tasks like segmentation, detection, and rendering

Neural Rendering based on One-Point-One NeRF (OPONeRF)

Instructed by Prof. Yueqi Duan, Department of Electronic Engineering at Tsinghua University 2023.05-2023.09

- Proposed OPONeRF to address the limitations of Neural Radiance Fields (NeRF) in handling scene changes
- Introduced a point-to-point NeRF approach for improved stability and rendering accuracy in dynamic scenes
- Developed a variational inference method to overcome computational and spatial resource challenges
- Conducted independent experiments, including perturbation tests, cross-scene rendering, and validation on simulated datasets

PUBLICATIONS

- **Hongru Yan***, Yu Zheng*, Yueqi Duan. Gaussian-Det: Learning Closed-Surface Gaussians for 3D Object Detection. ICLR2025 Accepted https://arxiv.org/abs/2410.01404 https://openreview.net/forum?id=DtFCIfvAFc
- **Hongru Yan**, Zeyuan Chen, Fangyin Wei, Xiang Zhang, Zhuowen Tu. Gaussian Swaying: A Surface-Oriented Framework for Wind-Driven Dynamics with 3D Gaussians. ICCV 2025 (Under Review)
- Yu Zheng, **Hongru Yan**, Yueqi Duan, Jiwen Lu. ShadowNeRF: Learning Neural Radiance Field with Sight Degradation and Recovery, TMM Accepted.
- Yu Zheng, Yueqi Duan, Kangfu Zheng, **Hongru Yan**, Jiwen Lu. OPONeRF: One-Point-One NeRF for Robust Neural Rendering. IJCV. (Under Review) https://arxiv.org/abs/2409.20043

COMPETITIONS

Electronic Design Competition (Tsinghua University, 3rd Prize)	2023.04
The 14th Mathematics competition of Chinese College Students (Chinese Mathematical Society, 1st Prize)	2022.12
The 8th China Undergraduate Physics Experiment Competition (Organizing Committee of CUPEC,2nd Prize)	2022.10
Mechanical and Electrical Design Competition (Department of Precision Instrument at THU,3rd Prize)	2021.12

SERVICES

Conference Review: ICLR 2025	
Associate Chair of Technological Innovation Department of Weixian College at Tsinghua University	2024.01-present
Head of Technological Innovation Department of Weixian College at Tsinghua University	2023.09-2024.01
Class League Secretary	2023.09-present
Student Union Officer at Tsinghua University	2021.09-2023.08
Officer of the science and technology association oftheDepartment of Precision Instrument at THU	2021.09-2023.08

OTHERS

- **Computer skills**: Python, Pytorch, Matlab, Origin, C++, C
- **Language skills**: Mandarin, English (TOEFL 106, GRE 331)